

Program 6

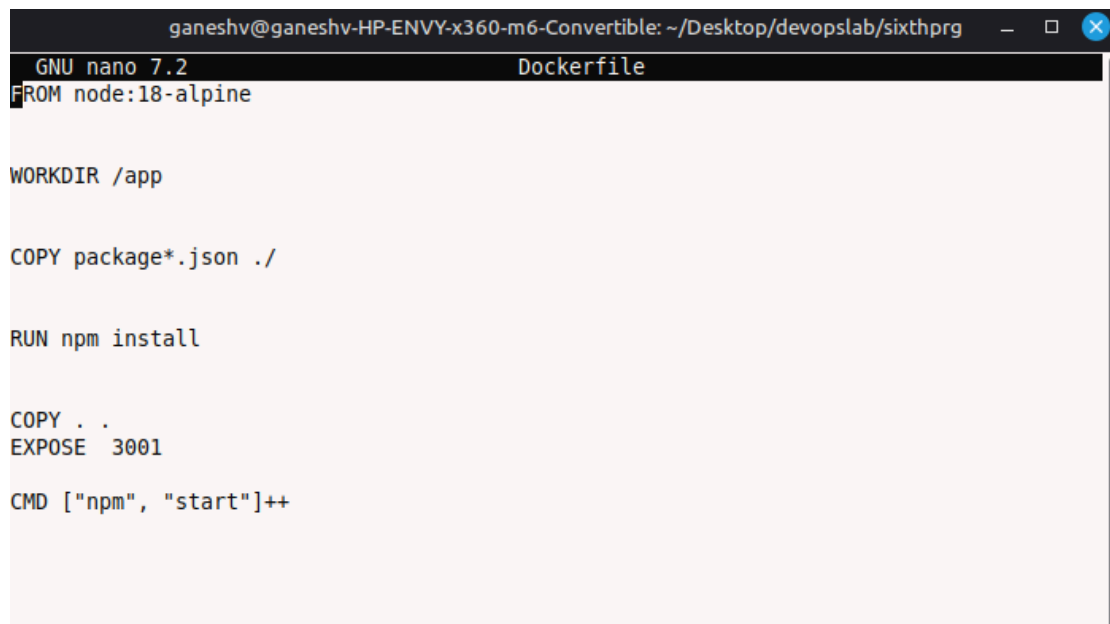
Docker Image Build, Push to Docker Hub & GitHub Container Registry

1. Create project directory

```
mkdir program-6 && cd program-6
```

Step 1: Create a Dockerfile

```
nano Dockerfile
```

A screenshot of a terminal window. The title bar shows the user 'ganeshv' on a machine named 'ganeshv-HP-ENVY-x360-m6-Convertible' at the path '~/Desktop/devopslab/sixthprg'. The terminal is running the 'nano' editor to edit a file named 'Dockerfile'. The content of the Dockerfile is as follows:

```
FROM node:18-alpine

WORKDIR /app

COPY package*.json ./

RUN npm install

COPY . .
EXPOSE 3001

CMD ["npm", "start"]++
```

Step 2: Build the Docker Image

```
docker build -t program-6 .
```

```

ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/sixthprg$ sudo docker build -t sixthprg .
[+] Building 2.5s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 161B                               0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine  2.3s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 5.66kB                                 0.0s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498 0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY package*.json ./                             0.0s
=> CACHED [4/5] RUN npm install                                    0.0s
=> [5/5] COPY . .                                                 0.1s
=> exporting to image                                             0.0s
=> => exporting layers                                             0.0s
=> => writing image sha256:b7a4afe78f2ca0d3303bef538bc7bba665bd07b3479dbe0dec3 0.0s
=> => naming to docker.io/library/sixthprg                        0.0s

1 warning found (use docker --debug to expand):
- JSONArgsRecommended: JSON arguments recommended for CMD to prevent unintended behavior related to OS si
gnals (line 16)
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/sixthprg$

```

Step 3: Tag the Image for Docker Hub

docker tag program-6 <dockerhub-username>/program-6:latest

```

ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/sixthprg$ docker tag sixthprg ganeshv2003/
program-6:latest
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/sixthprg$

```

Step 4: Push the Image to Docker Hub

Login to Docker Hub using your Docker Hub credentials (Personal Access Token as password).

docker login -u <dockerhub-username>

Push the image to Docker Hub:

docker push <dockerhub-username>/program-6:latest

```
ganeshv@ganeshv-HP-ENVY-x360-m6-Convertible:~/Desktop/devopslab/sixthprg$ docker login -u ganeshv2003
```

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password: █

Step 5: Tag the Image for GitHub Container Registry (GHCR)

Tag the same image for GitHub Container Registry.

```
docker tag program-6 ghcr.io/<github-username>/program-6:latest
```

Step 6: Push the Image to GHCR

Login to GHCR using GitHub Personal Access Token (PAT).

```
echo $GITHUB_PAT | docker login ghcr.io -u <github-username> --password-stdin
```

Push the image:

```
docker push ghcr.io/<github-username>/program-6:latest
```

Step 7: Connect the Local Project to GitHub (Source Code)

```
git init
git remote add origin https://github.com/<github-username>/program-6.git
git remote set-url origin
https://github.com/<github-username>/program-6.git git remote -v
```

Step 8: Push the Source Code to GitHub

git add .

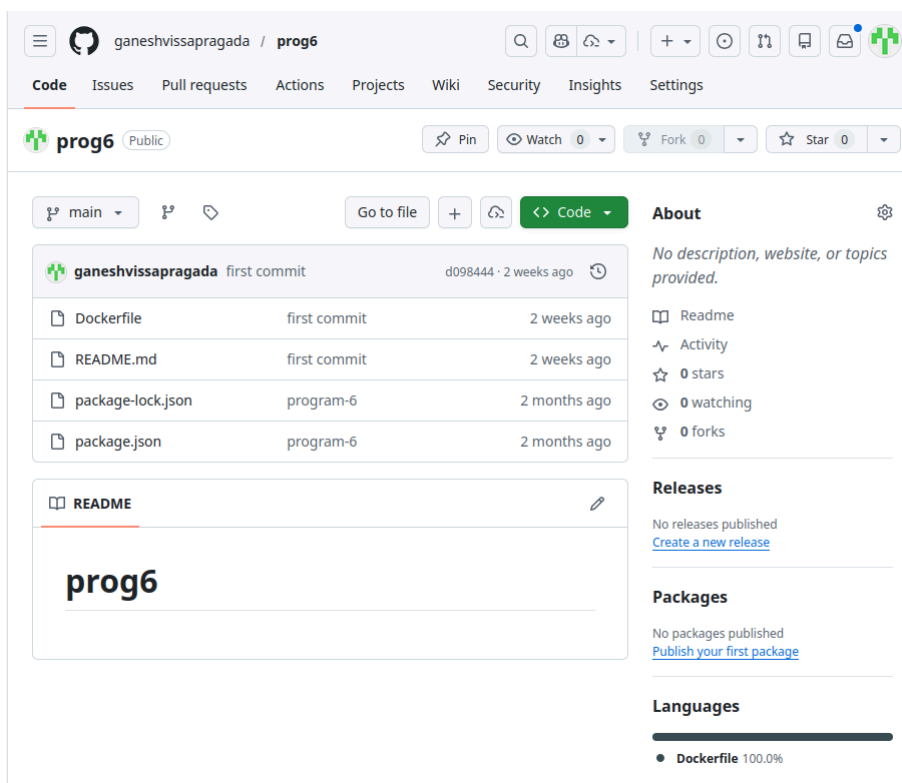
git commit -m "Initial commit"

git push -u origin main

Step 9: Verify Uploads on Both Registries

GitHub

- Ensure the repository program-6 exists on GitHub.
- Navigate to Packages.



Docker Hub

- Login to Docker Hub.
- Go to **Repositories**.

 ganeshv2003

Docker Personal

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Repositories

All repositories within the ganeshv2003 namespace.

 Search by repository name

All content

Create a repository

Name	Last Pushed 	Contains	Vis
ganeshv2003/program6	11 days ago	IMAGE	Put
ganeshv2003/ubuntu	about 2 months ago	IMAGE	Put

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